

## Mean Squared Error (MSE) / L2 Loss

- Q:** What is the primary role of a loss function in training ML models?  
**A:** It provides a quantitative measure of model performance by indicating the deviation between true and predicted values.
  - Q:** What is the mathematical formula for the Mean Squared Error ( $L_{\text{MSE}}$ )?  
**A:** ( $L_{\text{MSE}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$ )
  - Q:** Name two alternative terms used to refer to the Mean Squared Error (MSE).  
**A:** Quadratic Loss or L2 Loss
  - Q:** Why is MSE widely used in machine learning?  
**A:** Due to its mathematical simplicity and easy differentiability.
  - Q:** How does MSE's penalty for larger errors compare to smaller errors, and what is the consequence?  
**A:** Larger errors are penalized more strongly (quadratically), making MSE sensitive to outliers.
  - Q:** Minimizing the MSE is statistically equivalent to obtaining the Maximum Likelihood Estimation (MLE) parameters under what specific noise assumption?  
**A:** Gaussian (Normal) noise distribution.
  - Q:** For linear models, what mathematical property of the MSE function guarantees that the local minimum is also the global minimum?  
**A:** Convexity of the MSE function.
- 

## Mean Absolute Error (MAE) / L1 Loss

- Q:** What is the alternative name for the Mean Absolute Error (MAE)?  
**A:** L1 Loss
- Q:** What is the mathematical formula for the Mean Absolute Error ( $L_{\text{MAE}}$ )?  
**A:** ( $L_{\text{MAE}} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$ )
- Q:** In terms of outlier presence, how does the L1 Loss (MAE) compare to the L2 Loss (MSE)?  
**A:** L1 Loss (MAE) is more robust to outliers than L2 Loss (MSE).

11. **Q:** What is the main disadvantage of MAE (L1 Loss) when compared to MSE (L2 Loss) during gradient-based optimization?  
**A:** The derivative discontinuity of the absolute value function at zero can complicate gradient calculation.
12. **Q:** What is the relationship between Root Mean Squared Error (RMSE) and the target variable's unit?  
**A:** RMSE has the same unit as the target variable.
- 

## Mean-Squared Logarithmic Error (MSLE) and MAPE

13. **Q:** In which specific situations is Mean-Squared Logarithmic Error (MSLE) most applicable?  
**A:** When the target variable varies exponentially or spans several orders of magnitude.
14. **Q:** Due to the use of a logarithmic scale in MSLE, how are underestimations treated compared to overestimations?  
**A:** Underestimations are penalized more heavily than overestimations.
15. **Q:** What is the function of the "+1" terms added to  $y_i$  and  $\hat{y}_i$  in the MSLE formula?  
**A:** They allow logarithm calculation for zero values, ensuring all non-negative targets are valid.
16. **Q:** Minimizing MSLE is derived from the Maximum Likelihood Estimation (MLE) assuming what kind of noise distribution?  
**A:** Log-normal noise distribution.
17. **Q:** What kind of deviation (absolute or relative) does Mean Absolute Percentage Error (MAPE) use, and how is the error expressed?  
**A:** Relative deviation, expressed as a fraction or percentage of the true value.
- 

## Logistic Loss / Binary Cross-Entropy

18. **Q:** What are two alternative names for the Logistic Loss function?  
**A:** Log Loss and Binary Cross-Entropy Loss

19. **Q:** Which specific ML problem type and algorithm primarily utilize Logistic Loss?  
**A:** Logistic regression for binary classification problems.
20. **Q:** In the context of Logistic Loss, what does the predicted value  $\hat{y}_i$  represent?  
**A:** Predicted probability of the positive class ( $y=1$ ).
21. **Q:** Logistic Loss naturally arises from the Maximum Likelihood Estimation (MLE) of which specific probability distribution?  
**A:** Bernoulli distribution.
22. **Q:** What is the fundamental difference between Softmax Loss and Logistic Loss (Binary Cross-Entropy)?  
**A:** Softmax Loss is for multiclass classification; Logistic Loss is for binary classification.
23. **Q:** If the categorical cross-entropy (Softmax Loss) is used for a problem with only two classes (0 and 1), what does the formula reduce to?  
**A:** Binary cross-entropy (Log Loss).
- 

## Tweedie Loss and Other Loss Functions

24. **Q:** For what specific characteristic of data is the Tweedie Loss Function primarily used?  
**A:** Skewed data with a mix of low and continuous positive values.
25. **Q:** Give two real-world examples of data suitable for modeling with a Tweedie Loss Function.  
**A:** Rainfall amounts and energy consumption.
26. **Q:** How does the Tweedie parameter ( $\vartheta$ ) change the resulting loss function?  
**A:** Defines the family of loss functions:  $\vartheta=0 \rightarrow$  MSE,  $\vartheta=2 \rightarrow$  Gamma deviance.
27. **Q:** How is the Median Absolute Error (MedAE) calculated mathematically?  
**A:** Median of absolute differences  $|y_i - \hat{y}_i|$  across all data points.
28. **Q:** What does Max Error measure?  
**A:** Maximum absolute error among all data points.
29. **Q:** For complex models where analytical solutions are difficult, what numerical optimization technique is typically used?  
**A:** Stochastic Gradient Descent (SGD) or its variants.

30. **Q:** What is the fundamental relationship between maximizing the likelihood ( $\mathcal{L}$ ) and minimizing a loss function like MSE or Log Loss?

**A:** Maximizing likelihood is equivalent to minimizing negative log-likelihood (the loss function).